

A brief overview of Deep Unfolded Fractional Programming-Based Beamforming in RIS-Aided MISO Systems

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I. SYSTEM MODEL

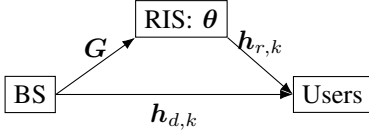


Fig. 1. A base concept of the system.

The model is described by a downlink MISO system. It consists of three main elements: the Base Station (BS), the users, and the Reconfigurable Intelligent Surface (RIS). The system has the following parameters:

- BS has M antennas.
- There are K users to be serviced.
- RIS has N elements, which causes a phase shift of $\theta \in \mathbb{C}^N$.
- $\mathbf{h}_{d,k} \in \mathbb{C}^M$ is the channel matrix between the BS and the user k .
- $\mathbf{G} \in \mathbb{C}^{N \times M}$ is the channel matrix between the BS and the RIS.
- $\mathbf{h}_{r,k} \in \mathbb{C}^N$ is the channel matrix between the RIS and the user k .

Therefore the user k receives the signal:

$$y_k = (\mathbf{h}_{d,k}^H + \mathbf{h}_{r,k}^H \text{diag}(\theta) \mathbf{G}) \sum_{i=1}^K \mathbf{v}_i s_i + n_k \quad (1)$$

from the BS. Where:

- s_i is data symbol for user i
- n_k add. white Gaussian noise. $n_k \sim \mathcal{CN}(0, \sigma^2)$
- $\sum_{i=1}^K \mathbf{v}_i s_i = \mathbf{x}$ the signal send by the the BS.
- $\mathbf{v}_i \in \mathbb{C}^{M \times 1}$ is the beamforming vector from the BS to the user i .
- $\mathbf{h}_{d,k}^H$ is direct path between the BS and user k .
- $\mathbf{h}_{r,k}^H \text{diag}(\theta) \mathbf{G}$ is the path from the BS to RIS, application of the phase shift and then the path form the RIS to the user k .
- Summary: $y_k = (\mathbf{h}_{dir}^H + \mathbf{h}_{indir}^H) \mathbf{x} + n_k$

The SINR can be calculated as:

$$\alpha_k = \frac{\left| \left(\mathbf{h}_{d,k}^H + \mathbf{h}_{r,k}^H \text{diag}(\theta) \mathbf{G} \mathbf{v}_k \right) \right|^2}{\sum_{i=1, i \neq k}^K \left| \left(\mathbf{h}_{d,k}^H + \mathbf{h}_{r,k}^H \text{diag}(\theta) \mathbf{G} \mathbf{v}_i \right) \right|^2 + \sigma_0^2} \quad (2)$$

which is simply the expected signal divided by the sum of the noise and all other signals sent.

The objective is to maximize the weighted sum rate (WSR):

$$(\mathcal{P}_1) : \max_{\mathbf{V}, \theta} \sum_{k=1}^K \omega_k \log(1 + \alpha_k) \quad (3)$$

$$\text{s.t. } |\theta_n| = 1, \forall n, \quad (4)$$

$$\sum_{k=1}^K \|\mathbf{v}_k\|^2 \leq P_{\max} \quad (5)$$

The system allows changing the beamforming vectors of the BS and the phase shift values of the RIS elements. $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_K]$ and ω_k a weight coefficient for the user k .

This is a typical multi-ratio fractional programming problem, with a possibly non-convex solution space.

II. BLOCK COORDINATE DESCENT METHOD

The Block Coordinate Descent method is an iterative optimization algorithm to solve multi-variable problems by partitioning the variables into smaller blocks. At each step, one variable is optimized while keeping the others fixed.

The optimization problem $(\mathcal{P}_1)$ is non-convex in two variables: the beamforming matrix $\mathbf{V}$ and the RIS phase-shift vector θ . A direct joint optimization is therefore not feasible.

First, the Lagrangian dual transform is applied to transform $\mathcal{P}_1$ into a more manageable form [1].

$$(\mathcal{P}_{1-1}) : \max_{\mathbf{V}, \theta, \gamma} \sum_{k=1}^K \omega_k \left[\log(1 + \gamma_k) - \gamma_k + \frac{(1 + \gamma_k) \alpha_k}{1 + \alpha_k} \right] \quad (6)$$

$$\text{s.t. } |\theta_n| = 1, \forall n, \quad (7)$$

$$\sum_{k=1}^K \|\mathbf{v}_k\|^2 \leq P_{\max}. \quad (7)$$

For fixed $(\mathbf{V}, \theta)$, the optimal auxiliary value is $\gamma_k^* = \alpha_k$. When γ is fixed, $(\mathcal{P}_{1-1})$ collapses to

$$(\mathcal{P}_2) : \max_{\mathbf{V}, \theta} \sum_{k=1}^K \frac{\omega_k (1 + \gamma_k) \alpha_k}{1 + \alpha_k}. \quad (8)$$

A. Update of $\mathbf{V}$

Let $\mathbf{h}_k = \mathbf{h}_{d,k} + \mathbf{G}^H \text{diag}(\theta) \mathbf{h}_{r,k}$

$$f_1(\mathbf{V}) = \sum_{k=1}^K \frac{\omega_k (1 + \gamma_k) |\mathbf{h}_k^H \mathbf{v}_k|^2}{\sum_{i=1}^K |\mathbf{h}_k^H \mathbf{v}_i|^2 + \sigma_0^2}, \quad (9)$$

Applying the quadratic transform [1] ...

$$f'_1(\mathbf{V}, \mathbf{z}) = \sum_{k=1}^K 2\sqrt{\omega_k(1+\gamma_k)} \operatorname{Re}\{z_k^* \mathbf{h}_k^H \mathbf{v}_k\} - \sum_{k=1}^K |z_k|^2 \left(\sum_{i=1}^K |\mathbf{h}_k^H \mathbf{v}_i|^2 + \sigma_0^2 \right). \quad (10)$$

$$z_k^\circ = \frac{\sqrt{\omega_k(1+\gamma_k)} \mathbf{h}_k^H \mathbf{v}_k}{\sum_{i=1}^K |\mathbf{h}_k^H \mathbf{v}_i|^2 + \sigma_0^2}. \quad (11)$$

$$\mathbf{v}_k^\circ = \sqrt{\omega_k(1+\gamma_k)} z_k (\eta \mathbf{I}_M + \mathbf{A})^{-1} \mathbf{h}_k, \quad (12)$$

where $\mathbf{A} = \sum_{k=1}^K |z_k|^2 \mathbf{h}_k \mathbf{h}_k^H$.

$$\eta^\circ = \min \left\{ \eta \geq 0 : \sum_{k=1}^K \|\mathbf{v}_k(\eta)\|_2^2 \leq P_{\max} \right\}, \quad (13)$$

B. Update of θ

define

$$\mathbf{a}_{i,k} = \operatorname{diag}(\mathbf{h}_{r,k}^H) \mathbf{G} \mathbf{v}_i, \quad b_{i,k} = \mathbf{h}_{d,k}^H \mathbf{v}_i, \quad (14)$$

the objective function ($\mathcal{P}_2$) can be rewritten as

$$f_2(\theta) = \sum_{k=1}^K \frac{\omega_k(1+\gamma_k) |b_{k,k} + \theta^H \mathbf{a}_{k,k}|^2}{\sum_{i=1}^K |b_{i,k} + \theta^H \mathbf{a}_{i,k}|^2 + \sigma_0^2}. \quad (15)$$

quadratic transform:

$$f'_2(\theta) = \sum_{k=1}^K 2\sqrt{\omega_k(1+\gamma_k)} \operatorname{Re}\{z_k^* \theta^H \mathbf{a}_{k,k} + z_k^* b_{k,k}\} - \sum_{k=1}^K |z_k|^2 \left(\sum_{i=1}^K |b_{i,k} + \theta^H \mathbf{a}_{i,k}|^2 + \sigma_0^2 \right), \quad (16)$$

where z_k is updated using ... then, the phase optimazation can be expressed as

$$f_3(\theta) = -\theta^H \mathbf{U} \theta + 2 \operatorname{Re}\{\theta^H \boldsymbol{\nu}\} + \chi, \quad (17)$$

where χ is independent of θ and

$$\mathbf{U} = \sum_{k=1}^K |z_k|^2 \sum_{i=1}^K \mathbf{a}_{i,k} \mathbf{a}_{i,k}^H, \quad (18)$$

$$\boldsymbol{\nu} = \sum_{k=1}^K \left(\sqrt{\omega_k(1+\gamma_k)} z_k^* \mathbf{a}_{k,k} - |z_k|^2 \sum_{i=1}^K b_{i,k}^* \mathbf{a}_{i,k} \right). \quad (19)$$

the update rule for θ is given by

$$\theta = \arg \min_{\theta} f'_3(\theta) \triangleq \theta^H \mathbf{U} \theta - 2 \operatorname{Re}\{\theta^H \boldsymbol{\nu}\}. \quad (20)$$

Due to constraints in (4), we substitute $\theta_n = e^{j\varphi_n}$ and obtain

$$\varphi = \arg \min_{\varphi \in \mathbb{R}^N} f'(\varphi) \triangleq (e^{j\varphi})^H \mathbf{U} e^{j\varphi} - 2 \operatorname{Re}\{\boldsymbol{\nu}^H e^{j\varphi}\}, \quad (21)$$

which is a non-convex problem whose optimal solution is hard to find. Following [1], we resort to the successive convex approximation (SCA) technique.

The full BCD method based on the Lagrangian dual transform thus updates $\mathbf{z}$, $\mathbf{V}$, φ , and $\boldsymbol{\alpha}$ repeatedly until convergence. In each outer iteration, two inner iterative processes

are also required. The BCD method is therefore a two-layer iterative process that relies on complex operations such as matrix inversion and one-dimensional search, leading to high computational complexity and latency. To deal with this issue, the DUFPP scheme is proposed in the next section.

REFERENCES

- [1] K. Shen and W. Yu, "Fractional programming for communication systems—part ii: Uplink scheduling via matching," *IEEE Transactions on Signal Processing*, vol. 66, no. 10, pp. 2631–2644, 2018.